# Alumni Fundraising Project

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**Summary**:
In this project, I will conduct a general exploratory analysis on sample university alumni gift data to reveal trends in the gifts received and the types of donors who have given them. I will also use recency, frequency, and monetary (RFM) analysis to determine which donors my hypothetical Development and Alumni Relations department should prioritize with their outreach.

**Data**:
The dataset used for this project was generated by AI. I asked ChatGPT to provide code in Postgresql that would generate a relational database that fits the context of a Business Intelligence Analyst position. To do this, I used keywords from a job listing as well as general information about Vanderbilt.

Here is part of my prompt: “The role is within the Development & Alumni Relations department of a T20 research university. The job summary specifically mentions that I'll be working with "fundraising and advancement" data.”

For more information on the relational database I used, please see the “AI Sample Database Tables - Snapshot” document.

*Please Note: While my dataset was generated with AI, **I certify that none of the code used in this project is AI generated**.

**Analysis:**
Below are the results of my SQL queries and insights. Please see my SQL script in the “alumni_fundraising_sample_script.sql” file.

–total all time gifts and the average gift amount

```
Total Gift Amount All Time|Average All Time Gift Amount|
--------------------------+----------------------------+
             $8,697,050.97|                     $434.85|
```

Insight: The university has received over $8.5 million in donations overall, and the average amount given by a donor is around $430.

–date range of gifts

| Date of Earliest Donation | Date of Most Recent Donation |
|---|---|
| 2016-03-18 | 2026-03-14 |

Insight: This dataset is reporting on around 10 years of donations.

–total gift amounts broken down by the type of gift

| Type of Gift | Total Gift Amount |
|---|---|
| Recurring | $1,916,755.19 |
| OneTime | $1,740,485.52 |
| Planned | $1,727,332.51 |
| Major | $1,695,362.08 |
| Pledge | $1,617,115.67 |

Insight: Recurring gifts make up the largest portion of the total amount of gifts given.

–total gift amounts broken down by the major of the alumni donors

| Major | Total Gift Amount |
|---|---|
| Engineering | $1,355,514.05 |
| English | $1,010,816.43 |
| Physics | $968,363.70 |
| Statistics | $945,983.43 |
| History | $921,881.23 |
| Computer Science | $908,385.29 |
| Chemistry | $889,195.59 |
| Economics | $884,197.87 |
| Biology | $780,541.66 |
| Finance | $25,614.17 |
| CS | $4,728.23 |
| Mechanical Engineering | $1,172.68 |
| Political Science | $656.64 |

Insight: Alumni who majored in Engineering have given the most in gifts, followed by English and Physics.

–total gift amounts broken down by the degree conferred

| Degree | Total Gift Amount |
|---|---|
| BA | $2,184,345.42 |
| MBA | $1,761,368.29 |
| PhD | $1,664,080.03 |
| BS | $1,617,070.45 |
| MS | $1,470,186.78 |

–total gift amounts broken down by the age of the alumni donors

| Age | Total Gift Amount | Number of Gifts |
|---|---|---|
| 36 | $584,814.56 | 651 |
| 53 | $348,508.01 | 705 |
| 40 | $319,345.25 | 602 |
| 25 | $305,642.22 | 522 |
| 32 | $280,428.59 | 621 |
| 52 | $274,734.90 | 616 |
| 50 | $271,198.81 | 649 |
| 48 | $265,357.42 | 608 |
| 27 | $252,912.53 | 593 |
| 38 | $251,007.54 | 597 |
| 51 | $247,105.30 | 632 |
| 43 | $246,356.95 | 577 |
| 47 | $244,386.44 | 525 |
| 55 | $244,142.91 | 596 |
| 35 | $241,730.12 | 620 |
| 37 | $240,244.49 | 555 |
| 41 | $239,687.49 | 611 |
| 22 | $239,390.02 | 536 |
| 30 | $231,256.51 | 611 |
| 28 | $229,147.31 | 587 |
| 23 | $225,795.17 | 604 |
| 39 | $225,693.32 | 538 |
| 34 | $223,072.99 | 538 |
| 29 | $222,544.87 | 509 |
| 42 | $219,274.43 | 556 |
| 44 | $217,597.67 | 521 |
| 56 | $210,577.83 | 523 |
| 54 | $209,551.46 | 601 |
| 24 | $208,802.62 | 559 |
| 49 | $206,327.21 | 510 |
| 26 | $202,475.89 | 507 |
| 33 | $197,200.54 | 538 |
| 31 | $192,285.41 | 519 |
| 45 | $185,082.04 | 499 |
| 46 | $167,757.98 | 461 |
| 61 | $25,614.17 | 3 |

–extracting the min, max, and count of ages (from the previous results)

| min | max | count |
|---|---|---|
| 22 | 61 | 36 |

Insight: The age range of alumni donors spans from age 22 to age 61.

–top 10 ages based on the total amount of gifts given

| Age | Total Gift Amount |
|---|---|
| 25 | $305,642.22 |
| 27 | $252,912.53 |
| 32 | $280,428.59 |
| 36 | $584,814.56 |
| 38 | $251,007.54 |
| 40 | $319,345.25 |
| 48 | $265,357.42 |
| 50 | $271,198.81 |
| 52 | $274,734.90 |
| 53 | $348,508.01 |

Insight: The top 10 ages in terms of total amount of gifts given ranges from 25 to 53, with donors aged 36 having given the most in gifts.

–total amount of gifts since the beginning of the year (01/01/2026)

| Total Gift Amount YTD |
|---|
| $1,282,221.81 |

Insight: We have received over $1.2 million in gifts since the beginning of the year.

–average number of months since last gift

| Average Number of Months Since Last Gift |
|---|
| 12 |

Insight: The average alumni donor has given their last gift about a year ago.

–total gift amounts broken down by the year given

| Year | Total Gift Amount |
|---|---|
| 2016 | $393,853.65 |
| 2017 | $511,978.95 |
| 2018 | $439,889.97 |
| 2019 | $453,984.38 |
| 2020 | $491,254.05 |
| 2021 | $579,517.52 |
| 2022 | $623,958.43 |
| 2023 | $864,841.90 |
| 2024 | $1,104,811.12 |
| 2025 | $1,942,742.46 |
| 2026 | $1,290,218.54 |

Insight: There is generally an increasing trend in the amount of gifts received per year. We received our all-time high in gifts last year (in 2025).

–breakdown of total gifts received in 2025 and 2026 looking at month by month

| Year | Month | Total Gift Amount |
|---|---|---|
| 2025 | 1 | $135,780.17 |
| 2025 | 2 | $88,300.08 |
| 2025 | 3 | $121,125.66 |
| 2025 | 4 | $119,486.73 |
| 2025 | 5 | $134,565.25 |
| 2025 | 6 | $116,434.97 |
| 2025 | 7 | $162,458.64 |
| 2025 | 8 | $173,268.94 |
| 2025 | 9 | $170,247.16 |
| 2025 | 10 | $168,566.03 |
| 2025 | 11 | $297,925.51 |
| 2025 | 12 | $254,583.32 |
| 2026 | 1 | $276,790.57 |
| 2026 | 2 | $389,069.02 |
| 2026 | 3 | $624,358.95 |

Insight: If you look at the first 3 months of 2025 and 2026, we have already received more in gifts this year than we did at this time last year. Assuming we stay on this trend, we can predict that we will earn more than $1.9 million in gifts this year.

**Were the campaigns successful?**
–total gift amounts broken down by campaign

| Campaign | Total Gift Amount | Target Amount | Campaign End Date |
|---|---|---|---|
| Annual Giving FY26 | $7,415,873.75 | $2,000,000.00 | 2026-06-30 |
| Campaign for Science Building | $30,000.00 | $50,000,000.00 | 2027-12-31 |
| Scholarship Drive 2025 | $100.00 | $500,000.00 | 2025-12-31 |
| | $1,251,077.22 | | |

Insight: The Annual Giving FY26 campaign is already successful a few months ahead of time. We haven't yet reached our goal for the Campaign for Science Building, but we still have until the end of next year. Unfortunately, we were unable to reach our goal of $500k for Scholarship Drive 2025.

**Who are the top donors?**
—top 5 donors by gift amount

| constituent_id | first_name | last_name | Total Gift Amount |
|---|---|---|---|
| 27 | First17 | Last17 | $336,777.44 |
| 2683 | First2653 | Last2653 | $100,656.15 |
| 1826 | First1796 | Last1796 | $94,114.45 |
| 2156 | First2126 | Last2126 | $55,962.72 |
| 202 | First172 | Last172 | $52,623.23 |

Insight: Donor First17 Last17 has given the most in gifts with over $336,000k given over his lifetime.

–demographic information for those 5 donors

| first_name | last_name | age | degree | grad_year | major | employer | job_title | Number of Donations | Total Gift Amount |
|---|---|---|---|---|---|---|---|---|---|
| First17 | Last17 | 36 | BA | 1994 | Engineering | University | Analyst | 63 | $336,777.44 |
| First172 | Last172 | 53 | MBA | 2002 | Statistics | ConsultingCo | Manager | 7 | $52,623.23 |
| First1796 | Last1796 | 40 | MBA | 2012 | English | ConsultingCo | Analyst | 8 | $94,114.45 |
| First2126 | Last2126 | 32 | BA | 1994 | Physics | BioLabs | Manager | 4 | $55,962.72 |
| First2653 | Last2653 | 25 | BA | 2006 | Engineering | University | Manager | 7 | $100,656.15 |

Insight: First 17 Last 17 is 36 years old, so they're likely skewing the data and making the total gift amount given by 36 year olds higher. Also, that person has given 63 gifts over the course of their life, which is much higher than the other top donors.

Other general findings: 2 out of the 5 donors majored in Engineering, and 3 out of the 5 donors are in manager positions.

**Recency, Frequency, and Monetary (RFM) Analysis**
This is the process of assigning three values to each donor based on their recency, frequency, and monetary scores, to help inform which donors my hypothetical Alumni Relations and Development department should prioritize.

**Recency (Number of days since last gift given)**
–percentiles for the recency analysis

| tenth_percentile | thirtieth_percentile | median | seventieth_percentile | nintieth_percentile |
|---|---|---|---|---|
| 3 | 28 | 102 | 299 | 1058 |

–After assigning a value 1-6 to each donor based on the percentile they fall in, here are the first 10 rows of that table. A value of 6 means they donated very recently (less than 3 days ago), and a value of 1 means they donated a long time ago (more than 2 years ago).

| constituent_id | days_since_last_gift | recency_value |
|---|---|---|
| 1489 | 894 | 2 |
| 4790 | 969 | 2 |
| 273 | 730 | 2 |
| 3936 | 1037 | 2 |
| 2574 | 161 | 3 |
| 951 | 254 | 3 |
| 4326 | 90 | 4 |
| 2614 | 918 | 2 |
| 2520 | 32 | 4 |
| 2466 | 60 | 4 |

**Frequency (The number of gifts given)**
–percentiles for the frequency analysis

| tenth_percentile | thirtieth_percentile | median | seventieth_percentile | nintieth_percentile |
|---|---|---|---|---|
| 2 | 3 | 4 | 5 | 7 |

–After assigning a value 1-6 to each donor based on the percentile they fall in, here are the first 10 rows of that table. A value of 6 means they have given a large number of gifts (more than 7), and a value of 1 means they have only given 1 gift (less than 2).

| constituent_id | count_gifts | frequency_value |
|---|---|---|
| 1489 | 5 | 4 |
| 4790 | 6 | 5 |
| 273 | 4 | 3 |
| 3936 | 2 | 1 |
| 2574 | 3 | 2 |
| 951 | 2 | 1 |
| 4326 | 7 | 5 |
| 2614 | 1 | 1 |
| 2520 | 6 | 5 |
| 2466 | 5 | 4 |

**Monetary (The total dollar amount of gifts given)**
–percentiles for the monetary analysis

| tenth_percentile | thirtieth_percentile | median | seventieth_percentile | nintieth_percentile |
|---|---|---|---|---|
| 367.72 | 739.25 | 1093.61 | 1600.35 | 4107.65 |

–After assigning a value 1-6 to each donor based on the percentile they fall in, here are the first 10 rows of that table. A value of 6 means they have given a larger dollar amount in gifts (more than ~$4100), and a value of 1 means they have given a smaller dollar amount in gifts (less than ~$360).

| constituent_id | sum_gifts | monetary_value |
|---|---|---|
| 1489 | 1934.77 | 5 |
| 4790 | 1994.47 | 5 |
| 273 | 974.38 | 3 |
| 3936 | 277.35 | 1 |
| 2574 | 637.08 | 2 |
| 951 | 1817.87 | 5 |
| 4326 | 2554.63 | 5 |
| 2614 | 357.13 | 1 |
| 2520 | 1963.97 | 5 |
| 2466 | 1246.44 | 4 |

–After combining all of these results into a new table that has four columns, the donor's ID and a column for each of their scores, we can filter for the donors who have 6's in all three categories. Here is the result of that query.

| constituent_id | recency_value | frequency_value | monetary_value |
|---|---|---|---|
| 4723 | 6 | 6 | 6 |
| 4254 | 6 | 6 | 6 |
| 32 | 6 | 6 | 6 |
| 2277 | 6 | 6 | 6 |
| 4784 | 6 | 6 | 6 |
| 3707 | 6 | 6 | 6 |
| 2072 | 6 | 6 | 6 |
| 1253 | 6 | 6 | 6 |
| 4923 | 6 | 6 | 6 |
| 490 | 6 | 6 | 6 |
| 1826 | 6 | 6 | 6 |
| 625 | 6 | 6 | 6 |
| 2158 | 6 | 6 | 6 |
| 327 | 6 | 6 | 6 |
| 1175 | 6 | 6 | 6 |
| 2442 | 6 | 6 | 6 |
| 1344 | 6 | 6 | 6 |

Insight: There are 17 donors who have given a large number of gifts, and high value of gifts, and have given a gift recently. Essentially, these are our most engaged and supportive donors.

–the demographic information of these 17 donors

| constituent_id | first_name | last_name | age | degree | grad_year | major | employer | job_title |
|---|---|---|---|---|---|---|---|---|
| 490 | First460 | Last460 | 22 | BS | 2009 | History | RetailCo | Manager |
| 625 | First595 | Last595 | 23 | MBA | 1990 | Engineering | University | Manager |
| 1253 | First1223 | Last1223 | 27 | BA | 2014 | Computer Science | BioLabs | Researcher |
| 3707 | First3677 | Last3677 | 31 | BS | 2012 | Computer Science | FinanceGroup | Director |
| 2277 | First2247 | Last2247 | 35 | MS | 2010 | Physics | ManufacturingCo | Researcher |
| 2442 | First2412 | Last2412 | 37 | MS | 2009 | Computer Science | ManufacturingCo | Owner |
| 4723 | First4693 | Last4693 | 39 | PhD | 2013 | Economics | BioLabs | Manager |
| 4923 | First4893 | Last4893 | 39 | MS | 1998 | Statistics | FinanceGroup | Consultant |
| 1826 | First1796 | Last1796 | 40 | MBA | 2012 | English | ConsultingCo | Analyst |
| 2072 | First2042 | Last2042 | 40 | PhD | 2004 | Computer Science | University | Director |
| 32 | First2 | Last2 | 45 | BS | 2004 | History | FinanceGroup | Manager |
| 2158 | First2128 | Last2128 | 46 | PhD | 1990 | History | FinanceGroup | Consultant |
| 1175 | First1145 | Last1145 | 48 | PhD | 2010 | Statistics | TechCorp | Analyst |
| 1344 | First1314 | Last1314 | 48 | BA | 2005 | Engineering | RetailCo | Owner |
| 4254 | First4224 | Last4224 | 49 | MS | 2013 | Economics | University | Analyst |
| 4784 | First4754 | Last4754 | 55 | MBA | 1994 | Physics | TechCorp | Engineer |
| 327 | First297 | Last297 | 56 | BA | 2012 | English | ManufacturingCo | Owner |

Insight: This group ranges in age from 22 to 56, with most of them being in their 30s and 40s. This could signal that alumni in their 30s and 40s are more likely to be engaged and supportive donors. Also, more than half of the donors in this group have some sort of leadership role, whether that be manager, director, or owner. Therefore, it's likely that targeting donors in those types of roles would be best.

**Conclusion:**

Overall, my hypothetical Development and Alumni Relations department is doing very well. We have been accepting gifts for 10 years and we've generally been receiving more in gifts each year. Although we weren't able to reach our goal for our campaign last year, we're crushing the Alumni Giving FY26 one for this year, and since we're projected to do better this year than last year, it's likely that we'll crush the Campaign for Science Building one, ending next year, too.

The RFM analysis revealed 17 donors who we should prioritize our outreach to. It also revealed trends in the types of donors that might fall into this group in the future. These types of donors are typically in their 30s and 40s, and have some sort of leadership role (e.g. manager, director, owner).

If you took a look at the schema diagram for this relational database, you noticed that there are additional tables that I didn't use in this project. Those tables simply didn't have enough data to be realistic, and I didn't want this report to be too long. If I were to build upon my work, I would expand the other tables, and look for trends related to who attended our events and then subsequently donated, which events garnered the most in gifts, which interactions/communications with donors were the most successful, and if there is a pattern between donors who engage/support us the most and where they live.